

– SAFETY DATA SHEET

NN-CalNit Calcium Nitrate

Industrial-Grade Calcium Nitrate Tetrahydrate

Prepared in the format of GHS / the Chemicals (MSIHC) Rules

Document No.	Revision	Issue Date	Product Code
NNPL/SDS/PROD-07	1.0	1 July 2026	NN-CalNit

01 IDENTIFICATION

Field	Detail
Product name	NN-CalNit Calcium Nitrate
Chemical name	Calcium nitrate tetrahydrate, $\text{Ca}(\text{NO}_3)_2 \cdot 4\text{H}_2\text{O}$
CAS number	13477-34-4 (tetrahydrate); 10124-37-5 (anhydrous)
Recommended use	Industrial oxidising salt – concrete admixture, wastewater, latex, fertigation
Available forms	Solid ($\geq 98\%$); 60% concentrated liquid solution (200 kg drums)
Supplier	NilayNarayan Polychem LLP, Plot D-167 (P), Kandra Ind. Area, Govindpur, Jharkhand 826010
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02 HAZARD IDENTIFICATION

GHS classification: Oxidising Solid, Category 3 (H272); Eye Irritation Category 2A (H319). Calcium nitrate intensifies fire; it is an oxidiser.

- **Oxidiser** – H272 – May intensify fire; oxidiser.
- **Eyes** – H319 – Causes serious eye irritation.
- **Irritant** – May cause mild skin and respiratory-tract irritation (dust).
- **Ingestion** – Harmful if swallowed in quantity; may cause methaemoglobinaemia via nitrate at very high intake.

Precautionary: keep away from heat and combustibles; wear eye protection and gloves; avoid dust; keep out of reach of children.

03 COMPOSITION / INFORMATION ON INGREDIENTS

Ingredient	CAS No.	Conc.	Remark
Calcium nitrate tetrahydrate	13477-34-4	$\geq 98\%$	Oxidiser; active substance (solid grade)
Calcium nitrate (in liquid grade)	13477-34-4	$\approx 60\%$	Active substance in aqueous solution
Water of crystallisation	7732-18-5	Balance	Bound (tetrahydrate)
Impurities (chloride, sulphate, iron, Pb)	—	Trace	Within specification limits

04 FIRST-AID MEASURES

- **Inhalation** — Move to fresh air. If dust has been inhaled and irritation persists, seek medical advice.
- **Skin** — Wash skin with soap and water; remove contaminated clothing. If irritation persists, seek medical advice.
- **Eyes** — Rinse cautiously with water for several minutes; remove contact lenses if easy; continue rinsing. Seek medical attention.
- **Ingestion** — Rinse mouth, drink water; do not induce vomiting. Obtain medical attention if a large amount is swallowed.

05 FIRE-FIGHTING MEASURES

- **Media** — Use plenty of water / water spray. The product itself is non-combustible but supports combustion of other materials.
- **Hazards** — As an oxidiser it can intensify a fire and release oxygen; may cause reignition. Cool exposed containers with water.
- **Protection** — Thermal decomposition releases nitrogen oxides (NOx). Wear self-contained breathing apparatus.

06 ACCIDENTAL RELEASE MEASURES

- **Precautions** — Wear gloves, goggles and dust protection. Keep away from combustible materials.
- **Containment** — Sweep or vacuum the solid into clean, dry containers; avoid dust generation. Do not return spilt material to original packaging.
- **Environment** — Prevent large releases into drains and watercourses (nitrate load). Flush residues with plenty of water.

07 HANDLING & STORAGE

- **Handling** — Avoid dust; use with adequate ventilation; wear protective equipment; wash after handling.
- **Storage** — Store cool, dry and well-ventilated; keep containers tightly closed to prevent moisture pickup and caking.
- **Segregation** — Segregate from combustibles, organics, reducing agents, acids and ammonium compounds (oxidiser).

08 EXPOSURE CONTROLS / PERSONAL PROTECTION

Control	Recommendation
Nitrate dust	No specific OEL; control as nuisance/inert dust — observe local limits
Respiratory protection	Dust mask (P1/P2) where dust is generated
Hand protection	Rubber / nitrile gloves
Eye protection	Safety goggles
Engineering controls	Local exhaust / general ventilation; eyewash station at handling point

09 PHYSICAL & CHEMICAL PROPERTIES

Property	Value
Appearance	White crystalline / granular solid; or clear-to-pale 60%

Property	Value
	liquid solution
Odour	Odourless
pH (5 % solution)	5.0 – 7.5
Melting point	≈ 42.7 – 44 °C (solid)
Specific gravity – solid	≈ 1.82
Density – 60% liquid solution	1.3 – 1.4 kg/l
Solubility in water	Highly soluble (~1200 g/l at 20 °C)
Oxidising properties	Oxidiser (supports combustion)

10 STABILITY & REACTIVITY

- **Stability** – Stable under normal conditions; hygroscopic – cakes if exposed to moisture.
- **Avoid** – Heat, contact with combustibles, reducing agents, strong acids and organic materials.
- **Decomposition** – Decomposes on strong heating to release oxygen and nitrogen oxides (NOx).
- **Hazardous reactions** – Can form explosive/violently-reactive mixtures with fuels, powdered metals and organics.

11 TOXICOLOGICAL INFORMATION

- **Irritation** – Irritant to eyes; mild irritant to skin and respiratory tract (dust).
- **Acute** – Low acute toxicity; large oral doses may cause gastrointestinal upset and, via nitrate, methaemoglobinaemia.
- **Chronic** – No chronic hazard expected under normal industrial handling with good hygiene.

12 ECOLOGICAL INFORMATION

Highly water-soluble; contributes nitrate to the environment. Large releases may cause nutrient enrichment (eutrophication) of surface waters. Prevent uncontrolled entry into drains, soil and watercourses.

13 DISPOSAL CONSIDERATIONS

Dispose of surplus product and contaminated packaging through authorised waste handlers in accordance with local regulations. Small quantities may be diluted with large volumes of water where permitted. Do not dispose of near combustible waste.

14 TRANSPORT INFORMATION

Field	Detail
UN number	UN 1454
Proper shipping name	Calcium nitrate
Transport hazard class	5.1 (Oxidising substance)
Packing group	III
Labels	Oxidiser (5.1)

15 REGULATORY INFORMATION

Classified and labelled as an oxidising solid under GHS. Industrial-grade product; observe applicable chemical-safety, storage and transport regulations for oxidisers, including the Manufacture, Storage and Import of Hazardous Chemicals (MSIHC) Rules.

16 OTHER INFORMATION

This SDS is provided for safe-handling guidance. Classification and exposure limits must be verified against the regulations in force at the place of use. Issued by the NilayNarayan Polychem LLP NABL-accredited laboratory.

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